

Part 1 - Overview, Architecture, Authentication and Roles

Mini ERP + CRM Operations Portal study notes for understanding the assignment foundation.

How to read this pack

Blue boxes are explicit case-study requirements recovered from the source explanation. Green boxes are implementation recommendations or inferences. This distinction helps you explain what was mandatory and what you chose as a developer.

Language style: simple Hinglish, interview-focused, and implementation-oriented. The goal is to help you build the project and confidently explain why each part exists.

1. Project Ki Core Understanding

Case-study requirement

Build a Mini ERP + CRM Operations Portal for a wholesale/distribution company. It should combine customers, products, inventory, sales challans, accounts-related visibility, follow-ups, and role-based access in one internal web app.

Simple language mein: company ke employees abhi customers, stock, sales aur follow-ups alag-alag places par manage kar rahe honge, jaise Excel, WhatsApp ya manual notes. Project ka idea hai ek single internal portal banana jahan business operations organized rahen.

```
MINI ERP + CRM
Customers -> CRM follow-ups
Products -> Inventory and stock
Sales -> Challans and stock reduction
Users -> Role-based permissions
Database -> Single source of truth
```

ERP ka matlab business operations manage karna: products, stock, sales, challans. CRM ka matlab customer relationship manage karna: customer details, status, follow-up date, notes. Assignment mein dono ka combination expected hai.

2. Actors and Roles

Case-study requirement

Required roles: Admin, Sales, Warehouse and Accounts. JWT-based authentication is acceptable.

Role	Simple meaning	Likely access
Admin	System ka owner/super user	Users, customers, products, inventory, challans, reports - everything
Sales	Customer aur sales challan handle karta hai	Customers, follow-up notes, create challans, view own sales
Warehouse	Stock aur product quantity maintain karta hai	Products, inventory, stock movements, low-stock alerts
Accounts	Sales/financial information verify karta hai	Confirmed challans, amounts, customer/account details

Interview mein bolna: role-based access isliye zaroori hai kyunki internal portal mein har employee ko sab kuch edit karne dena business risk hai. Warehouse ko customer follow-up edit karne ki zaroorat nahi, Sales ko user management ki zaroorat nahi.

3. Technology Stack

Case-study requirement

Backend: Node.js, TypeScript, Express.js or NestJS, PostgreSQL or MySQL, REST APIs, validation and error handling. Frontend: React with HTML/CSS/JavaScript or TypeScript and responsive UI.

Recommended implementation

For a 48-hour assignment, choose React + TypeScript + Node.js + Express + PostgreSQL. Prisma is a good ORM choice because schema, migrations and queries are easier to explain.

```
React Frontend
|
| HTTP / REST API
v
Node.js + Express + TypeScript
```

```
|
| SQL / ORM
v
PostgreSQL or MySQL
```

Frontend direct database se baat nahi karta. Frontend sirf backend APIs call karta hai. Backend validation, permissions, stock rules, transactions aur database updates handle karta hai. Ye separation production systems mein normal architecture hai.

4. Authentication Flow

Authentication ka first screen login page hoga. User email/password enter karega. Backend credentials verify karega. Agar valid hai, backend JWT token generate karega aur frontend us token ko future requests mein bhejega.

```
POST /auth/login
Request: { email, password }
Backend: verify password hash
Backend: generate JWT with user id and role
Response: { token, user }
Frontend: store token
Next API calls: Authorization: Bearer <JWT>
```

JWT Simple Explanation

JWT ek signed token hota hai. Iske andar user ka id aur role jaise claims store ho sakte hain. Backend token verify karke samajh leta hai ki request kis user se aa rahi hai. Password har request ke saath bhejne ki zaroorat nahi hoti.

5. Authorization and Route Protection

Authentication ka matlab user logged in hai. Authorization ka matlab user ko particular action karne ki permission hai ya nahi. Dono alag concepts hain.

Action	Check
View dashboard	JWT valid hona chahiye
Create customer	Sales/Admin role allowed
Update stock	Warehouse/Admin role allowed
Confirm challan	Sales/Admin role plus stock validation
Manage users	Only Admin

Interview point
Frontend route guards useful hain, but real permission checks backend middleware mein hone chahiye. User browser manipulate karke frontend checks bypass kar sakta hai.

6. Middleware Design

```
authMiddleware:
  read Authorization header
  verify JWT
  attach user to request

roleMiddleware(['ADMIN', 'SALES']):
  check request.user.role
  allow or return 403 Forbidden
```

7. What To Show In Demo

- Login with Admin, Sales, Warehouse and Accounts test users.
- Show that each role sees relevant menu items.
- Try a restricted action and explain that backend returns 403 Forbidden.
- Explain JWT token flow briefly using the Authorization header.
- Mention password hashing instead of plain text passwords.